Samarateja Reddy Gutta

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SUMMARY

Software engineer with 1.5 years of experience building backend systems and LLM applications. Built Portico, a live multi-tenant AI intake platform where a Claude agent books real appointments through strict-schema tools, and Access AI Gateway, a guardrail pipeline that makes LLM database access auditable. At YuppTV, built a Kafka and SQS event pipeline ingesting 500,000+ events per day and shared on-call for a platform with 10,000+ daily users. M.S. in Computer Science (Machine Intelligence and Cognition), George Washington University.

TECHNICAL SKILLS

Languages: Python, Java 21, TypeScript, JavaScript (Node.js), SQL, Bash, HTML/CSS

AI and LLMs: Claude API tool use (Anthropic SDK), Gemini API, OpenAI structured outputs, Model Context Protocol (MCP), LLM-as-judge evaluation, prompt-injection defense, PII masking, audit logging

Backend and APIs: FastAPI, Spring Boot, Spring Security (OAuth2 resource server), JWT, OAuth 2.0, Express, REST API design, gRPC/Protobuf, webhooks, third-party API integration, background workers

Databases: PostgreSQL, SQLite, Redis (cache), YugabyteDB YSQL, JPA/Hibernate, SQLAlchemy, Flyway, Alembic, CTEs, window functions, indexing, EXPLAIN ANALYZE, transactions, SELECT FOR UPDATE SKIP LOCKED

Distributed Systems and Messaging: Microservices, Apache Kafka, Amazon SQS, content-addressed storage, partial-failure design, idempotent retries, circuit breakers, single-flight caching, Raft basics

Cloud and DevOps: AWS (Lambda, S3, ECR, IAM, CloudWatch, SQS), Terraform, Docker, GitHub Actions, Vercel, Render, Linux

Testing: pytest, JUnit 5, MockMvc, Testcontainers, Vitest, integration and failure-mode tests, contract tests, mocked-client offline tests

EXPERIENCE

Software Engineer
YuppTV, Hyderabad, India | Jun 2023 – Aug 2024

- Shared the on-call rotation for a platform serving 10,000+ daily active users, triaging production issues from AWS CloudWatch dashboards, logs and automated alarms on microservice latency and error budgets, through to root cause and fix.
- Architected an event-driven streaming pipeline with Apache Kafka and Amazon SQS across microservices, ingesting 500,000+ daily telemetry events with fault-tolerant delivery and zero message loss.
- Decreased production defect recurrence by 40% by building automated CI/CD integration test suites with Docker, Testcontainers and GitHub Actions, expanding coverage from 55% to 88%.
- Reduced query bottlenecks on core high-volume tables by designing two-tier data access layers and optimizing complex SQL (CTEs, indexing, window functions), improving read throughput.
- Owned scoped tickets in Agile sprints from planning through implementation, testing and release.

Software Engineering Intern
HP / National University of Singapore, Singapore | Jun 2022 – Aug 2022

- Standardized relational schemas across 2M+ records by writing SQL ETL pipelines over 10+ disparate datasets using advanced joins, window functions and CTEs.
- Eliminated downstream model ingestion failures by building automated data validation with schema-contract enforcement, catching raw data anomalies before processing.
- Trained and evaluated baseline scikit-learn models to find key feature correlations across enterprise data.

PROJECTS

Portico — AI Client Intake and Scheduling Platform for Law Firms | Python, FastAPI, PostgreSQL, Next.js, TypeScript, Claude API, Twilio

- Built a multi-tenant platform for immigration law firms: Next.js and TypeScript staff dashboard and an embeddable chat widget where Claude runs client intake in English, Spanish and French (live at porticointake.com).
- Claude agent uses strict-schema tools (save_lead, find_slots, book_slot) to book consultations from real calendar availability; urgent matters email the on-call attorney.
- Enforced tenant isolation with a non-null firm_id on every firm table, with tests proving one firm gets a 404 on another firm's leads and appointments; Argon2 passwords, hashed session tokens, A-number/SSN/passport masking before storage.
- Booking re-checks the slot inside the transaction (409 on a race); reminder worker uses PostgreSQL SELECT FOR UPDATE SKIP LOCKED with retries; Twilio SMS with signed-webhook verification and STOP/HELP handling.

- 327 test functions in GitHub Actions CI; 30-conversation eval set (English/Spanish/French) graded by a Claude Haiku judge for legal-advice refusal, urgency and required fields.

Access AI Gateway — LLM Guardrails for Database Access | TypeScript, Node.js, Anthropic SDK, PostgreSQL, Vitest

- TypeScript/Node gateway around an Anthropic SDK tool-use loop that lets an LLM query a multi-tenant PostgreSQL database.
- Every tool call passes a known-tool check, role allowlist, tenant-scope enforcement, scoped SQL, result scan, untrusted-content wrapping with a per-call random delimiter, and an audit row (denials included).
- Prompt-injection detection with 7 scoring rules plus NFKC normalization, homoglyph mapping, zero-width stripping, base64 decode-and-rescan and split-payload scanning; 6 evasion cases tested and benign look-alikes pass.
- 32 Vitest tests; includes a script that disables one guardrail stage at a time to show exactly what leaks without it.

CSV Analyst Chat | React, FastAPI, pandas, OpenAI API

- Upload a CSV and ask questions in English; gpt-4o-mini returns a structured query plan (JSON schema), never code.
- Plans are validated against the real schema with one self-correction retry, then executed by pandas on a stateless FastAPI backend; client-side dashboard with delimiter detection, type inference and a trend chart; 12-question benchmark script.

Movie Catalog Service — MCP Server and REST API | Python, asyncio, httpx, FastAPI, MCP SDK

- One domain layer behind an MCP server (4 tools) and a FastAPI HTTP API; about 3,500 lines and 82 tests, with a live demo.
- Pooled httpx client with jittered exponential backoff honoring Retry-After, a circuit breaker, and a TTL cache with single-flight behavior (5 identical concurrent requests make 1 upstream call).
- Concurrent 4-call fan-out with graceful degradation, rate limiting, JSON logs with x-request-id, and a /health endpoint reporting breaker and cache state.

EDUCATION

M.S. Computer Science, Machine Intelligence and Cognition

The George Washington University, Washington, D.C. | *Completed May 2026*

Coursework: Machine Learning, Distributed Systems, Advanced Algorithms, Database Systems, Cloud Computing. Graduate research: EEG sleep event classification.

B.Tech Computer Science

Mahindra University, Hyderabad, India | *Completed May 2024*

Lab Assistant for the undergraduate Big Data course (Hadoop and Spark lab sessions).